

```

C:\MNOVA\ARM\ARM Technology\STM32_Embedded_Course\ARM_Programming_Course\Task_Projects\my_workspace
arm-none-eabi-gcc -mcpu=cortex-m4 -mthumb -stdgnu11 -Wall -O0 -g -ffastmath -c main.c -o main.o
arm-none-eabi-gcc -mcpu=cortex-m4 -mthumb -stdgnu11 -Wall -O0 -g -ffastmath -c led.c -o led.o
arm-none-eabi-gcc -mcpu=cortex-m4 -mthumb -stdgnu11 -Wall -O0 -g -ffastmath -c stm32_startup.o -o stm32_startup.o
arm-none-eabi-gcc -nostdlib -T stm32_1s.ld -Wl,-Map=final.map main.o led.o stm32_startup.o -o final.elf

C:\MNOVA\ARM\ARM Technology\STM32_Embedded_Course\ARM_Programming_Course\Task_Projects\my_workspace
arm-none-eabi-gdb /d:/grym Technology/STM32_Embedded_Course/ARM_Programming_Course/Task_Projects/my_workspace/final.elf
(gdb) make openocd
(gdb) openocd -f /d:/grym Technology/STM32_Embedded_Course/OpenOCD/0.10.0-12-20190422-2015/sr
OpenOCD v0.10.0-12-20190422-2015/sr
GNU MCU Eclipse OpenOCD, 64-bit/open On-Chip Debugger 0.10.0-dev-0593-g23d80f4 (2019-04-22-20:25)
(c) 2019 Open On-Chip Debugger Ltd.
For bug reports, read
http://openocd.org/doc/doxygen/bugs.html
Info: The selected transport took too long to level target control. The results might differ compared to plain JTAG
Info: adapter speed: 2000 kHz
adapter_srst_delay: 100
none separate
Info: already separate srst_nogate srst_open_drain connect_deassert_srst
Info: listening on port 6666 for tcp connections
Info: listening on port 4444 for telnet connections
Info: clock speed 2000 kHz
Info: stm32f4x: GPIO VDDIO 0483:3748
Info: Target voltage: 3.25049V
Info: stm32f4x.cpu: hardware has 6 breakpoints, 4 watchpoints
Info: listening on port 3333 for gdb connections
Info: accepting 'gdb' connection on tcp/3333
Info: device = '0x00000000'
Info: flash size = 512 kibibytes
Info: flash size = 512 bytes
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x00100000 pc: 0x08000644 msp: 0x20020000
Info: Unable to match requested speed 8000 kHz, using 4000 kHz
Info: Unable to match requested speed 8000 kHz, using 4000 kHz
adapter speed: 4000 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x00100000 pc: 0x08000644 msp: 0x20020000
Info: Unable to match requested speed 8000 kHz, using 4000 kHz
Info: Unable to match requested speed 8000 kHz, using 4000 kHz
adapter speed: 4000 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x00100000 pc: 0x08000644 msp: 0x20020000
Invalid subcommand 'write image erase final.elf'
In procedure 'Flash'
(gdb) monitor flash write_image erase final.elf
auto erase enabled
wrote 16384 bytes from file final.elf in 0.668315s (23.940 KiB/s)
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x00100000 pc: 0x08000644 msp: 0x20020000
Info: Unable to match requested speed 8000 kHz, using 4000 kHz
Info: Unable to match requested speed 8000 kHz, using 4000 kHz
adapter speed: 4000 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x00100000 pc: 0x08000644 msp: 0x20020000
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x00100000 pc: 0x08000644 msp: 0x20020000
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x00100000 pc: 0x08000644 msp: 0x20020000
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x00100000 pc: 0x08000644 msp: 0x20020000
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x00100000 pc: 0x08000644 msp: 0x20020000
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x00100000 pc: 0x08000644 msp: 0x20020000
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x00100000 pc: 0x08000644 msp: 0x20020000
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x00100000 pc: 0x08000644 msp: 0x20020000
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x00100000 pc: 0x08000644 msp: 0x20020000
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x00100000 pc: 0x08000644 msp: 0x20020000
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x00100000 pc: 0x08000644 msp: 0x20020000
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x00100000 pc: 0x08000644 msp: 0x20020000
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x00100000 pc: 0x08000644 msp: 0x20020000
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x00100000 pc: 0x08000644 msp: 0x20020000
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x00100000 pc: 0x08000644 msp: 0x20020000
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x00100000 pc: 0x08000644 msp: 0x20020000
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x00100000 pc: 0x08000644 msp: 0x20020000
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x00100000 pc: 0x08000644 msp: 0x20020000
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x00100000 pc: 0x08000644 msp: 0x20020000
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x00100000 pc: 0x08000644 msp: 0x20020000
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x00100000 pc: 0x08000644 msp: 0x20020000
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x00100000 pc: 0x08000644 msp: 0x20020000
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x00100000 pc: 0x08000644 msp: 0x20020000
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x00100000 pc: 0x08000644 msp: 0x20020000
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x00100000 pc: 0x08000644 msp: 0x20020000
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
Info: Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 18
```

integrating syscalls.c

MINGW64:/d/Argyan Technology/STM32_Embedded_Course/ARM_Programming_Course/Task_Project/my_workspace

```
Lenovo@Lenovo MINGW64 /d/Argyan Technology/STM32_Embedded_Course/ARM_Programming_Course/Task_Project/my_workspace (main)
$ make
arm-none-eabi-gcc -mcpu=cortex-m4 -mthumb -std=gnu11 -Wall -O0 -g -ffreestanding -c main.c -o main.o
arm-none-eabi-gcc -mcpu=cortex-m4 -mthumb -std=gnu11 -Wall -O0 -g -ffreestanding -c led.c -o led.o
arm-none-eabi-gcc -mcpu=cortex-m4 -mthumb -std=gnu11 -Wall -O0 -g -ffreestanding -c stm32_startup.c -o stm32_startup.o
arm-none-eabi-gcc -mcpu=cortex-m4 -mthumb -std=gnu11 -Wall -O0 -g -ffreestanding -c syscalls.c -o syscalls.o
arm-none-eabi-gcc --specs=nano.specs --specs=nosys.specs -T stm32_ls.ld -Wl,-Map=final.map main.o led.o stm32_startup.o syscalls.o -o final.elf
```

```
Lenovo@Lenovo MINGW64 /d/Argyan Technology/STM32_Embedded_Course/ARM_Programming_Course/Task_Project/my_workspace (main)
$ arm-none-eabi-objdump.exe -h final.elf
```

final.elf: file format elf32-littlearm

```
Sections:
Idx Name          Size      VMA       LMA       File off  Algn
0 .text           00003278  08000000  08000000  00010000  2**2
CONTENTS, ALLOC, LOAD, READONLY, CODE
1 .init           00000018  08003278  08003278  00013278  2**2
CONTENTS, ALLOC, LOAD, READONLY, CODE
2 .fini           00000018  08003290  08003290  00013290  2**2
CONTENTS, ALLOC, LOAD, READONLY, CODE
3 .eh_frame       00000004  080032a8  080032a8  000132a8  2**2
CONTENTS, ALLOC, LOAD, READONLY, DATA
4 .ARM.exidx      00000008  080032ac  080032ac  000132ac  2**2
CONTENTS, ALLOC, LOAD, READONLY, DATA
5 .data           00000070  20000000  080032b4  00020000  2**2
CONTENTS, ALLOC, LOAD, DATA
6 .init_array     00000004  20000070  08003324  00020070  2**2
CONTENTS, ALLOC, LOAD, DATA
7 .fini_array     00000004  20000074  08003328  00020074  2**2
CONTENTS, ALLOC, LOAD, DATA
8 .bss            00000090  20000078  0800332c  00020078  2**2
ALLOC
9 .ARM.attributes 0000002c  00000000  00000000  00020078  2**0
CONTENTS, READONLY
10 .comment       00000049  00000000  00000000  000200a4  2**0
CONTENTS, READONLY
11 .debug_info     00000d1b  00000000  00000000  000200ed  2**0
CONTENTS, READONLY, DEBUGGING, OCTETS
12 .debug_abbrev   000004e3  00000000  00000000  00020e08  2**0
CONTENTS, READONLY, DEBUGGING, OCTETS
13 .debug_aranges 00000080  00000000  00000000  000212eb  2**0
CONTENTS, READONLY, DEBUGGING, OCTETS
14 .debug_ranges   00000018  00000000  00000000  000213eb  2**0
CONTENTS, READONLY, DEBUGGING, OCTETS
15 .debug_line     0000077f  00000000  00000000  00021383  2**0
CONTENTS, READONLY, DEBUGGING, OCTETS
16 .debug_str      00000592  00000000  00000000  00021b02  2**0
CONTENTS, READONLY, DEBUGGING, OCTETS
17 .debug_frame    00001148  00000000  00000000  00022094  2**2
CONTENTS, READONLY, DEBUGGING, OCTETS
```

```
Lenovo@Lenovo MINGW64 /d/Argyan Technology/STM32_Embedded_Course/ARM_Programming_Course/Task_Project/my_workspace (main)
$ |
```

Semi hosting

```
MINGW64/d/Argyan Technology/STM32_Embedded_Course/ARM_Programming_Course/Task_Project/my_workspace
$ make opened
OpenOCD /f d:/Argyan Technology/STM32_Embedded_Course/ARM_Programming_Course/OpenOCD/0.10.0-12-20190422-2015/scr
ipts/board/st_nucleo_f4.cfg
GNU MCU ECLIPSE OpenOCD, 64-bitOpen On-Chip Debugger 0.10.0-dev-00593-g23ad80df4 (2019-04-22-20:25)
Licensed under GNU GPL v2
For bug reports, read
    http://openocd.org/doc/docgen/bugs.html
Info : The selected transport took over low-level target control. The results might differ compared to plain JTAG
/sdo
adapter speed: 2000 kHz
adapter_nsrst_delay: 100
none separate
srst_only separate srst_nogate srst_open_drain connect_deassert_srst
Info : Listening on port 6666 for tcl connections
Info : Listening on port 4444 for telnet connections
Info : clock speed 2000 kHz
Info : STLink V2.14.3 (part V2) VID:PID 0483:3748
Info : Target voltage: 3.224021
Info : stm32f4xx.cpu: hardware has 6 breakpoints, 4 watchpoints
Info : Listening on port 3333 for gdb connections
Info : accepting 'gdb' connection on tcp/3333
Info : device id = 0x10004a11
Info : flash size = 512 bytes
Info : flash size = 512 bytes
seminhosting is enabled
Info : Unable to match requested speed 2000 kHz, using 1800 kHz
Info : Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x01000000 pc: 0x080007cc msp: 0x20020000, seminhosting
Info : Unable to match requested speed 8000 kHz, using 4000 kHz
Info : Unable to match requested speed 8000 kHz, using 4000 kHz
adapter speed: 4000 kHz
Info : Unable to match requested speed 2000 kHz, using 1800 kHz
Info : Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x01000000 pc: 0x080007cc msp: 0x20020000, seminhosting
Info : Unable to match requested speed 8000 kHz, using 4000 kHz
Info : Unable to match requested speed 8000 kHz, using 4000 kHz
adapter speed: 4000 kHz
Info : Unable to match requested speed 2000 kHz, using 1800 kHz
Info : Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x01000000 pc: 0x080007cc msp: 0x20020000, seminhosting
Info : halted: PC: 0x080002b2

MINGW64/d/Argyan Technology/STM32_Embedded_Course/ARM_Programming_Course/Task_Project/my_workspace
$ make all_sh
arm-none-eabi-gcc -mcpu=cortex-m4 -mthumb -stdgnu11 -Wall -O0 -g -ffreestanding -c main.c -o main.o
arm-none-eabi-gcc -mcpu=cortex-m4 -mthumb -stdgnu11 -Wall -O0 -g -ffreestanding -c led.c -o led.o
arm-none-eabi-gcc -mcpu=cortex-m4 -mthumb -stdgnu11 -Wall -O0 -g -ffreestanding -c stm32_startup.c -o stm32_star
tup.o
arm-none-eabi-gcc -mcpu=cortex-m4 -mthumb -mfloat-abi=soft --specs=rdimon.specs -T stm32_ls.ld -Wl,-Map=final_sh.
map main.o led.o stm32_startup.o -lc -lrdimon -o final_sh.elf
$ make debug_sh
arm-none-eabi-gdb final_sh.elf \
    -ex "target remote localhost:3333" \
    -ex "monitor arm seminhosting enable" \
    -ex "monitor reset init" \
    -ex "load" \
    -ex "break main" \
    -ex "continue"
GNU gdb GDB Arm Embedded Toolchain 10.3-2021.103 10.2.90.20210621-git
Copyright (C) 2021 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software; you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
<http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from final_sh.elf...
Remote debugging using localhost:3333
main () at main.c:55
55      __asm volatile ("MSR MSP, NO" :: "r" (SCHED_STACK_START));
seminhosting is enabled
Unable to match requested speed 2000 kHz, using 1800 kHz
Unable to match requested speed 2000 kHz, using 1800 kHz
adapter speed: 1800 kHz
target halted due to debug-request, current mode: Thread
xPSR: 0x01000000 pc: 0x080007cc msp: 0x20020000, seminhosting
Unable to match requested speed 8000 kHz, using 4000 kHz
Unable to match requested speed 8000 kHz, using 4000 kHz
adapter speed: 4000 kHz
Loading section .text, size 0x9920 1ma 0x80000000
Loading section .eh_frame, size 0x4 1ma 0x80009920
Loading section .ARM.exidx, size 0x8 1ma 0x80009924
Loading section .data, size 0xadc 1ma 0x8000992c
Loading section .init_array, size 0x4 1ma 0x8000a408
Loading section .fini_array, size 0x4 1ma 0x8000a40c
Loading section .init_array.000000, size 0x4 1ma 0x8000a410
Start address 0x80000000, load size 42004
Transfer rate: 22 KB/sec, 4667 bytes/write.
Breakpoint 1 at 0x80000000: file main.c, line 55.
Note: automatically using hardware breakpoints for read-only addresses.
Continuing.
breakpoint 1, main () at main.c:55
55      __asm volatile ("MSR MSP, NO" :: "r" (SCHED_STACK_START));
(gdb) continue
Continuing.
halted: PC: 0x080002b2
```